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class NaiveBayes:
    def __init__(self):
        self.prior = {}
        self.conditional = {}
        self.classes = []

    def fit(self, data):
        total_records = len(data)
        labels = [row[-1] for row in data]
        self.classes = list(set(labels))

        # Calculate Priors
        for c in self.classes:
            self.prior[c] = labels.count(c) / total_records
            self.conditional[c] = {}

        # Calculate Conditionals
        num_features = len(data[0]) - 1
        for c in self.classes:
            sub_data = [row[:-1] for row in data if row[-1] == c]
            sub_total = len(sub_data)

            for f_idx in range(num_features):
                self.conditional[c][f_idx] = {}
                feature_vals = [row[f_idx] for row in sub_data]
                for val in set(feature_vals):

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                    count = sum(1 for row in sub_data if row[f_idx] == val)
                    # Using Laplace smoothing
                    self.conditional[c][f_idx][val] = (count + 1) / (sub_total +
len(set(feature_vals)))

    def predict(self, sample):
        best_class = None
        best_prob = -1.0

        for c in self.classes:
            prob = self.prior[c]
            for f_idx, val in enumerate(sample):
                if val in self.conditional[c][f_idx]:
                    prob *= self.conditional[c][f_idx][val]
            if prob > best_prob:
                best_prob = prob
                best_class = c
        return best_class

# Example Usage
if __name__ == "__main__":
    dataset = [
        ['Sunny', 'High', 'No'],
        ['Sunny', 'High', 'No'],
        ['Overcast', 'High', 'Yes'],
        ['Rainy', 'Normal', 'Yes'],
        ['Rainy', 'Normal', 'Yes']
    ]

    nb = NaiveBayes()
    nb.fit(dataset)
    test_instance = ['Sunny', 'Normal']
    print(f"Predicted class for {test_instance}:", nb.predict(test_instance))

```

## Program 9: Naive Bayes Classification (Second Run)

Assigned Labels: [0 0 0 1 1 1]

Calculated Silhouette Validation Score: 0.7133477791749615

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